TUT-5

26/09/20

1. PE → $u(x) = b(x^2-a^2)^2$, $b = 2\,J/m^4$
$a = 1\,m$

(i) for eqm pts —

$$\frac{du}{dx} = 0 \Rightarrow 2b(x^2-a^2)\cdot 2x = 0$$
$$\Rightarrow 4bx(x^2-a^2) = 0$$
$$\Rightarrow x = 0 \;\Big|\; x^2-a^2 = 0 \Rightarrow x = \pm a$$

So, 3 eqm pts. — $x = -a, 0, +a$.

Now,
$$\frac{d^2u}{dx^2} = 4b(x^2-a^2) + 4bx\cdot 2x$$
$$= 4b(x^2-a^2) + 8bx^2$$
$$= 12bx^2 - 4ba^2$$

$$\left.\frac{d^2u}{dx^2}\right|_{x=-a} = 12\cdot b\cdot a^2 - 4ba^2$$
$$= 8ba^2 = 8\cdot 2\cdot 1 = \cancel{10}\ 16 > 0$$

$$\left.\frac{d^2u}{dx^2}\right|_{x=0} = -4ba^2 = -4\cdot 2\cdot 1 = -8 < 0$$

$$\left.\frac{d^2u}{dx^2}\right|_{x=+a} = 12ba^2 - 4ba^2 = 16 > 0$$

Condn for stable eqm : $\frac{d^2u}{dx^2} > 0$

So, the system have stable eqm at $x = -a, a$.

(ii) $u(x) = b(x^2-a^2)^2$

→ $x = -a, 0, +a$ → 0 → unstable ~~(min. PE)~~; $-a, +a$ → stable

at $x = \pm a$, $u(x) = b(a^2-a^2)^2 = 0$

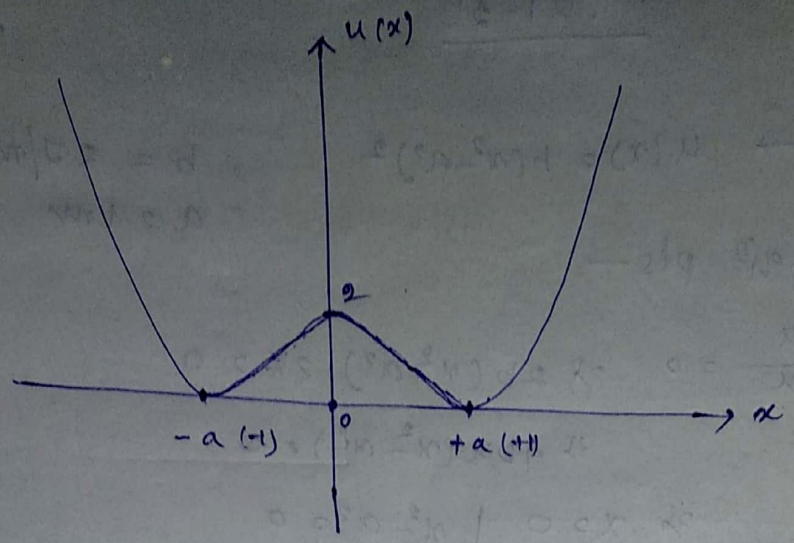
at $x = 0$, $u(x) = b\,a^4$
$= 2$.

$x = \pm a$ → min PE

at $x \to +\infty$,
$u(x) \to +\infty$

at $x \to -\infty$
$u(x) \to +\infty$

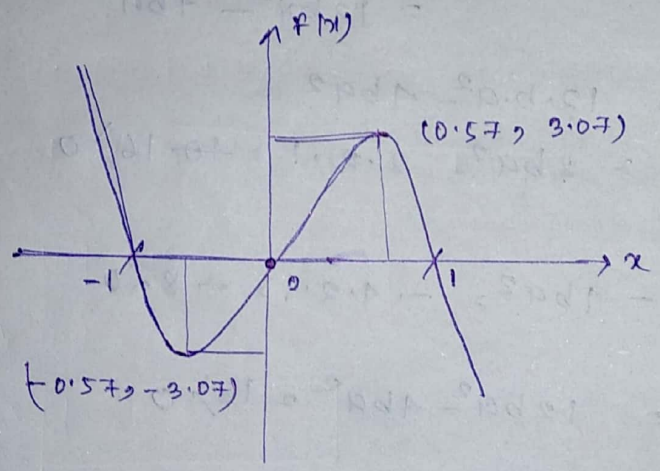
Scanned with CamScanner



(iii) $U(x) = b(x^2-a^2)^2$

$F_x = -\frac{\partial U}{\partial x} = -\left[2b(x^2-a^2).2x\right]$

$\Rightarrow F_x = -4bx(x^2-a^2)$



(iv) ~~$v_i = 4a$~~ $x_i = 4a = 4 \cdot 1 = 4$

initial velocity $v_i = 0$

$\therefore$ KE $(x = 4a) = 0 = K$

PE: $= U(x=4a) = b(16a^2-a^2)^2 = 225ba^4$

$= 450J.$

Total energy $(E) = K + U$

$= 0 + 450 = 450J$

Scanned with CamScanner

at $x=0$, $u(x=0) = b(-a^2)^2$
$= ba^4$
$= 2J$

$\therefore E = K(x=0) + U(x=0)$

$\Rightarrow 450 = K + 2$

$\Rightarrow K = 448J$

~~Now~~ $\Rightarrow \frac{1}{2}mv^2 = 448$

$\Rightarrow \boxed{v = \sqrt{\frac{2\times 448}{m}} = \sqrt{\frac{896J}{m}}}$

lik $x=-a$ $U(x=-a) = 0$

$\therefore E = K(x=-a) + U(x=-a)$
$= K + 0$

$\Rightarrow K = \cancel{448J}\ 450J$

$\therefore \Rightarrow \frac{1}{2}mv^2 = 450J$

$\Rightarrow \boxed{v = \sqrt{\frac{900J}{m}}}$

---

(2). $x = 10\sin(10t + \pi/3)$

general form $x = a\sin(\omega t + \phi)$

Comparing, $a = 10$ $\omega = 10$ $\phi = \pi/3$

(a) when, the particle will come to rest, it's velocity will be zero ($v=0$)

$x = 10\sin(10t + \pi/3)$

$\Rightarrow v = \frac{dx}{dt} = (10)^2\cos(10t + \pi/3)$.

So $v = 0$

$\Rightarrow \cos(10t + \pi/3) = 0$

$\Rightarrow (10t + \pi/3) = (2n+1)\pi/2$, ~~n=0, ±1, ±2, ±3, ...~~

with, $n = 0, \pm 1, \pm 2, \pm 3, \ldots$

The particle will first come to rest, when $n = 0$.

$\therefore 10t + \frac{\pi}{3} = \pi/2$

$\Rightarrow 10t = \frac{\pi}{6}$

$\Rightarrow \boxed{t = \frac{\pi}{60} \text{ sec}}$

(b) $x = 10 \sin(10t + \pi/3)$

$v = \frac{dx}{dt} = (10)^2 \cos(10t + \pi/3)$

$a = \frac{d^2x}{dt^2} = \left| -(10)^3 \sin(10t + \pi/3) \right|$

for, acc. to be zero,

$\sin(10t + \pi/3) = 0$

$\Rightarrow (10t + \pi/3) = n\pi, \quad n = 0, \pm 1, \pm 2, \ldots$

for, first zero acc. $n = 1$ [$n \neq 0$, then t becomes $-ve$]

$\therefore 10t + \pi/3 = \pi$

$\Rightarrow 10t = 2\pi/3$

$\Rightarrow \boxed{t = \pi/15 \text{ sec}}$

(c) $v = (10)^2 \cos(10t + \pi/3)$

for $v = max$ $\quad \cos(10t + \pi/3) = 1$

$\Rightarrow (10t + \pi/3) = n\pi, \quad n = 0, \pm 1, \pm 2, \ldots$

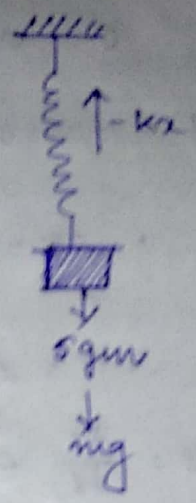
for, first v_{max}, $n = 1$, ($n \neq 0$, then $t \to -ve$)

$\therefore 10t + \pi/3 = \pi$

$\Rightarrow \boxed{t = \pi/15 \text{ sec}}$

Scanned with CamScanner

8. for spring-mass system-



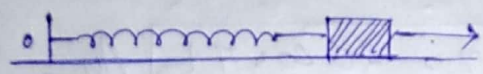
$F = -kx$

$\Rightarrow |F| = kx$

$\Rightarrow mg = kx$

$\Rightarrow k = \frac{mg}{x} = \frac{5 \times 980}{20}$

$\boxed{k = 245 \text{ dynes/cm}}$



pulled by 20 cm

(a) $F = -kx$

$\Rightarrow m\frac{d^2x}{dt^2} = -kx$

$\Rightarrow \frac{d^2x}{dt^2} + \frac{k}{m}x = 0$

$\Rightarrow \frac{d^2x}{dt^2} + \omega^2 x = 0$; with $\omega = \sqrt{\frac{k}{m}}$, angular freq^n

$= \sqrt{\frac{245}{5}} = \sqrt{49} = 7$

$\Rightarrow \boxed{\frac{d^2x}{dt^2} + 49x = 0}$ $\longrightarrow$ diff. eq^n for spring-mass system.

Initial conditions: at $t = 0$, $x = 20$ cm
$\Rightarrow v = \frac{dx}{dt} = 0$

(b) $\frac{d^2x}{dt^2} + 49x = 0$

$\Rightarrow \frac{d^2x}{dt^2} + \omega^2 x = 0$

$\Rightarrow (D^2 + \omega^2)x = 0$

with $D = \frac{d}{dt}$

Let solution, $x = e^{mt}$

auxiliary eq^n:

$m^2 + \omega^2 = 0$

$\Rightarrow m = \pm i\omega$

$\therefore x(t) = A\cos\omega t + B\sin\omega t$

Scanned with CamScanner

$x = A\cos\omega t + B\sin\omega t$

at $t=0$, $x = 20$ cm

$\therefore 20 = A$

$\therefore x = 20\cos\omega t + B\sin\omega t$

$\Rightarrow v = \frac{dx}{dt} = -20\omega\sin\omega t + B\omega\cos\omega t$

$\Rightarrow v|_{t=0} = 0 + B\omega = 0$

$\Rightarrow B = 0$

$\therefore \boxed{x = 20\cos\omega t} \Rightarrow \boxed{x = 20\cos 7t}$ Ans

AM Let $x = A\cos(\omega t + \phi)$

$\Rightarrow \frac{dx}{dt} = -A\omega\sin(\omega t + \phi)$

at $t=0$, $x = 20$

$\therefore 20 = A\cos\phi$

$\Rightarrow \cos\phi = \frac{20}{A}$

~~$\Rightarrow 20$~~

so, $\frac{20}{A} = \cos 0 = 1$

$\Rightarrow \boxed{A = 20}$

at $t=0$, $v=0$

$\Rightarrow \sin\phi = 0$

$\Rightarrow \boxed{\phi = 0}$

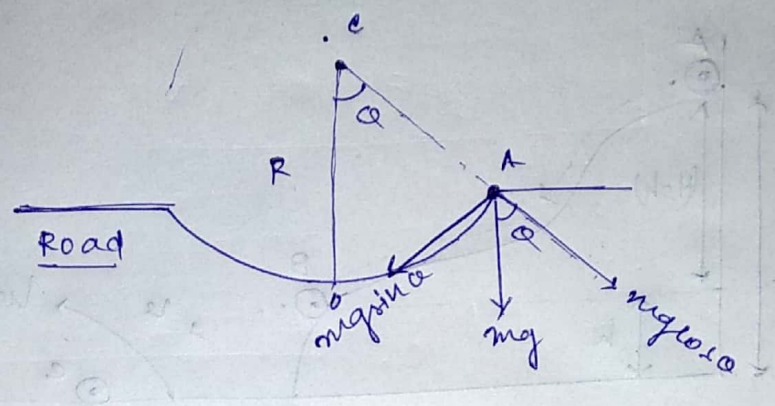
$\therefore x = A\cos\omega t$

$\Rightarrow \boxed{x = 20\cos 7t}$ Ans

Scanned with CamScanner

4. The bottom of a dip on a road has a radius R. A Rickshaw of mass m, left a little away from the bottom, oscillates about this dip. Deduce an expression for the time period of oscillation.

⇒



⇒ Let, R be the radius of the dip and 'C' be it's center.

As, shown in the figure, let the rickshaw of mass m be at 'A' at any instant.

The system is similar to the simple pendulum

The force that produces oscillation in the rickshaw is $F = mg\sin\theta \simeq mg\theta$

~~= mgsinθ~~ [[for θ being ~~be~~ very small]

Now, the displacement of rickshaw

= AO i.e the arc of the circle of radius R

⇒ $S = R\theta$

Now, for this simple pendulum type system

$F = KS$, $K \rightarrow$ force const.

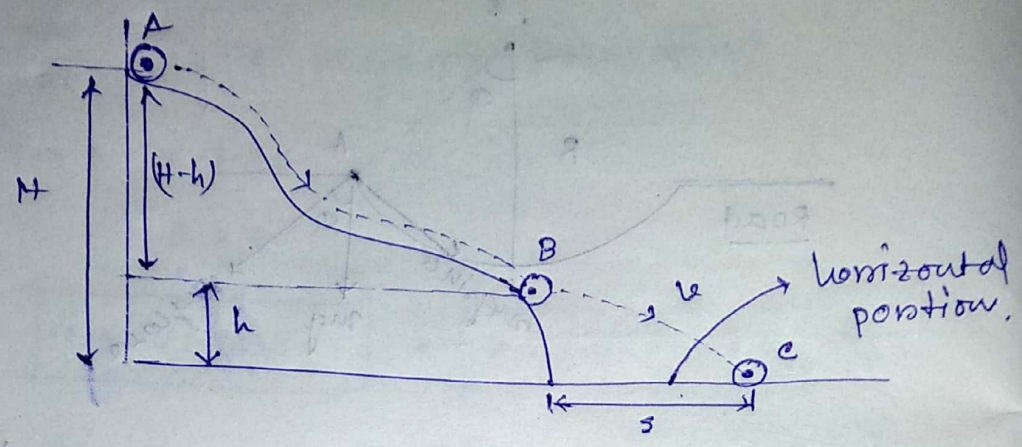
$\Rightarrow K = F/S = \frac{mg\theta}{R\theta} = \frac{mg}{R}$

So, time period of oscillation, $\boxed{T = 2\pi\sqrt{\frac{m}{K}} = 2\pi\sqrt{\frac{R}{g}}}$

Scanned with CamScanner

(82) A small disk A slides down with zero initial velocity from the top of a smooth hill of height H having a horizontal portion. What must be the height of the horizontal portion 'h' to ensure the max. distance covered by the disk?

⇒



⇒ The motion of the disk is as shown, followed by the path ABC.

The disk leaves the hill at the point 'B' and then lands at a point 'C' on its horizontal surface.

If, 'v' is the velocity of the disk at the portion B, ~~loss of~~ then,

Loss of PE of the disk in falling through a height (H-h) = gain in KE.

i.e, $mg(H-h) = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$

$= \frac{1}{2}mv^2 - 0$

$\Rightarrow v = \sqrt{2g(H-h)}$

If, 't' is the time taken by the disk to move from B to C (while falling through a height h) then,

$h = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2h}{g}}$

Scanned with CamScanner

Now $s = vt$

$$\Rightarrow s = \sqrt{2g(H-h)} \cdot \sqrt{2h/g}$$

$$= 2\sqrt{h(H-h)}$$

For, s to be ~~minimum~~, maximum.

$$\frac{ds}{dh} = 0$$

$$\Rightarrow \cancel{2} \cdot \frac{H-2h}{\cancel{2}\sqrt{h(H-h)}} = 0$$

$$\Rightarrow H = 2h$$

$$\Rightarrow h = H/2$$

$$\therefore s = 2\sqrt{\frac{H}{2}\left(H-\frac{h}{2}\right)} = 2\,\frac{H}{2}$$

$$\Rightarrow \boxed{s = H}$$

Scanned with CamScanner